LESSON 5.7

Determine Surface Area of Prisms

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the surface area of rectangular and triangular prisms.

WHY THIS MATTERS Your class is painting the outside of different time capsule shapes to protect them from weather. Some capsules are rectangular prisms and others are triangular prisms (wedge-shaped). To buy the right amount of paint, you need to calculate the surface area of each shape! One capsule is a rectangular prism 7 inches long, 4 inches wide, and 3 inches tall. Another is a triangular prism whose triangle bases are 8 inches wide and 3 inches tall, with slant sides of 5 inches, and the prism is 9 inches long.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Surface area *Spanish: Área de superficie*	The total area of all the flat sides of a solid.	add every face
Rectangular prism *Spanish: Prisma rectangular*	A solid box shape with six flat rectangle sides.	3 congruent pairs
Triangular prism *Spanish: Prisma triangular*	A solid with two triangle ends and three flat rectangle sides.	Shaped like a tent or a wedge of cheese — 2 triangle ends + 3 rectangle sides
Face *Spanish: Cara*	One flat side of a solid shape.	top, side, end
Net *Spanish: Plantilla (desarrollo plano)*	A flat shape that folds up into a solid.	unfolded box
Lateral face *Spanish: Cara lateral*	The side faces of a solid, not the top or bottom.	On a triangular prism, the 3 rectangles that wrap around the sides are lateral faces

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A closed box is 6 cm × 4 cm × 3 cm.

1. Faces: 6 × 4 = 24, 6 × 3 = 18, 4 × 3 = 12.
2. Double each: 48, 36, 24.
3. Add: 48 + 36 + 24 = 108.

Answer: SA = 108 cm²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a rectangular prism: 5 long, 4 wide, 3 tall. How many faces does it have?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The prism has _____ faces in _____ matching pairs.
- The three different face areas are _____ , _____ , and _____ .
- I doubled each face area because _____ .
- The surface area is _____ square units.

Word bank surface area | net | face | congruent | pair | prism | unfold | add | double | square units | top | side

WATCH OUT!

A common mistake

A common mistake in Surface Area of Prisms is confusing surface area with volume

LESSON 5.7

Determine Surface Area of Prisms

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 Write the name of the correct group on each line.

Put the steps in order for finding the surface area of a triangular prism.

Groups: Steps in order (first → last)

_____ Find the area of one triangular base

_____ Multiply the triangle area by 2 (two bases)

_____ Find the area of each rectangular lateral face

_____ Add all three rectangle areas together

_____ Add the two triangle areas + all rectangle areas = SA

HINT The cards describe finding a triangular prism's SA: triangles first, then rectangles, then a final total.

2 Write the letter of the matching item on each line.

Match each solid to its description.

_____ Rectangular prism

_____ Triangular prism

_____ Cube

_____ Surface area

_____ Volume

_____ Net

(A) 6 faces (all rectangles)

(B) 5 faces (2 triangles + 3 rectangles)

(C) 6 identical square faces

(D) Uses square units (in²)

(E) Uses cubic units (in³)

(F) 2D pattern that folds into 3D

HINT Read each card: some describe face counts, some describe units, and one describes a flat fold-out pattern.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A time capsule is a rectangular prism with l = 5 cm, w = 4 cm, h = 3 cm. What surface area needs paint?

(A) 47 cm²
(B) 60 cm²
(C) 94 cm²
(D) 94 cm³

HINT You need surface area — the total of all 6 face areas of the 5 × 4 × 3 prism — not the space inside.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

How many faces does a triangular prism have?

(A) 3
(B) 4
(C) 5
(D) 6

HINT Picture a tent or a triangular chocolate box: two triangle ends plus rectangles wrapping around the sides.

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

A wedge-shaped time capsule is a triangular prism. It has two triangular bases each with area 12 cm² and three rectangular faces with areas 30 cm², 40 cm², and 40 cm². What is the total surface area to paint?

(A) 110 cm²
(B) 122 cm²
(C) 134 cm²
(D) 134 cm³

HINT All five face areas are given: two triangles of 12 cm² each, and rectangles of 30, 40, and 40 cm².

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake — A Forgotten Face

1. **Dimensions** Rectangular prism time capsule: l = 5 in, w = 3 in, h = 2 in.
2. **Formula** SA = 2lw + 2lh + 2wh (add all 6 faces).
3. **Calculate** The student wrote SA = 2(5×3) + 2(5×2) + (3×2) = 30 + 20 + 6 = 56 in².

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT A prism has 6 faces, in 3 matching pairs — every pair gets multiplied by 2.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Compare Capsule A (rectangular, SA = 248 in²) and Capsule C (triangular, SA = 184 in²). Even though they are similar in size, why does the rectangular prism need more paint?

SENTENCE STARTER The rectangular prism has ___ faces while the triangular prism has ___ faces. The rectangular prism's SA is ___ in² which is ___ more than the triangular prism's ___ in² because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.7

Determine Surface Area of Prisms

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Calculate the surface area of each rectangular prism time capsule.

Capsule	Length	Width	Height	Surface Area
Capsule E	12 in	8 in	5 in	
Capsule F	9 cm	6 cm	6 cm	
Capsule G	15 ft	4 ft	3 ft	

SHOW YOUR WORK

2 **Use the model to solve. Show your work.**



What is the combined area of the left and right faces?

SHOW YOUR WORK

Answer: _____

3 **Write the name of the correct group on each line.**

Sort these prisms from LEAST surface area to GREATEST.

Groups: **Least → Greatest surface area**

_____ Cube 3×3×3: SA = 54

_____ Rect prism 5×4×3: SA = 94

_____ Rect prism 8×5×3: SA = 158

_____ Rect prism 10×6×4: SA = 248

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A wedge-shaped time capsule is a triangular prism. Its triangular bases have base 6 in and height 4 in. The prism is 10 in long and the two slant sides of the triangle are each 5 in. What surface area needs paint?

(A) 124 in²

(B) 160 in²

(C) 184 in²

(D) 184 in³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake — Surface Area or Volume?

1. **Dimensions** Rectangular prism time capsule: l = 7 in, w = 4 in, h = 3 in. Find the surface area.
2. **Formula** Surface area adds all 6 faces: SA = 2lw + 2lh + 2wh.
3. **Calculate** The student multiplied the three dimensions: 7 × 4 × 3 = 84, so they wrote SA = 84 in².

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Compare Capsule A (rectangular, SA = 248 in²) and Capsule C (triangular, SA = 184 in²). Even though they are similar in size, why does the rectangular prism need more paint?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 5.7

Determine Surface Area of Prisms

Challenge · Extension

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- ☐ Exceeds
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

A rectangular prism and a triangular prism have similar dimensions. Compare their surface areas and volumes.

Shape	Faces	Surface Area	Volume
Rect prism: 6×4×3	6		
Tri prism: base=6, h=4, length=3, sides=5	5		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Surface Area Error

1. **Dimensions** Rectangular prism: l = 12 cm, w = 7 cm, h = 5 cm
2. **Formula** SA = 2lw + 2lh + 2wh
3. **Calculate** SA = 2(12×7) + 2(12×5) + 2(7×5) = 168 + 120 + 35 = 323 cm²
4. **Answer** Surface Area = 323 cm²

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Compare two closed containers: a rectangular prism (8 × 6 × 4 in) and a triangular prism whose ends are right triangles with perpendicular legs 6 in and 8 in and hypotenuse 10 in. The triangular prism is 8 in long. Which uses less protective film to cover every face? Show your work.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Compare Capsule A (rectangular, SA = 248 in²) and Capsule C (triangular, SA = 184 in²). Even though they are similar in size, why does the rectangular prism need more paint?

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